RC CAR 중간 보고서 2

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필요 권한과 GATT

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필요 권한

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
<uses-permission android:name="android.permission.BLUETOOTH" />
<uses-permission android:name="android.permission.BLUETOOTH_ADMIN" />
<uses-permission android:name="android.permission.BLUETOOTH_SCAN" />
<uses-permission android:name="android.permission.BLUETOOTH_CONNECT" />
<uses-permission android:name="android.permission.BLUETOOTH_ADVERTISE" />
<uses-permission android:name="android.permission.POST_NOTIFICATIONS" />
```

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GATT 통신

계층적 구조로 데이터 구성.

- 서비스 : 기능 묶음. Ex) 심박수 서비스
- 특성 : 하나의 데이터 단위 Ex) 심박수 수치
- 디스크립터 : 특성의 추가 정보 Ex) 단위, 포맷

Serial 통신의 경우 데이터가 1byte씩 순차적으로 들어와 전송되지만 GATT 통신의 경우 데이터가 수십 바이트로 묶여서 전송됨.

BT05A는 저전력 블루투스 모듈인 BLE. BLE는 GATT 방식으로 통신해야함.

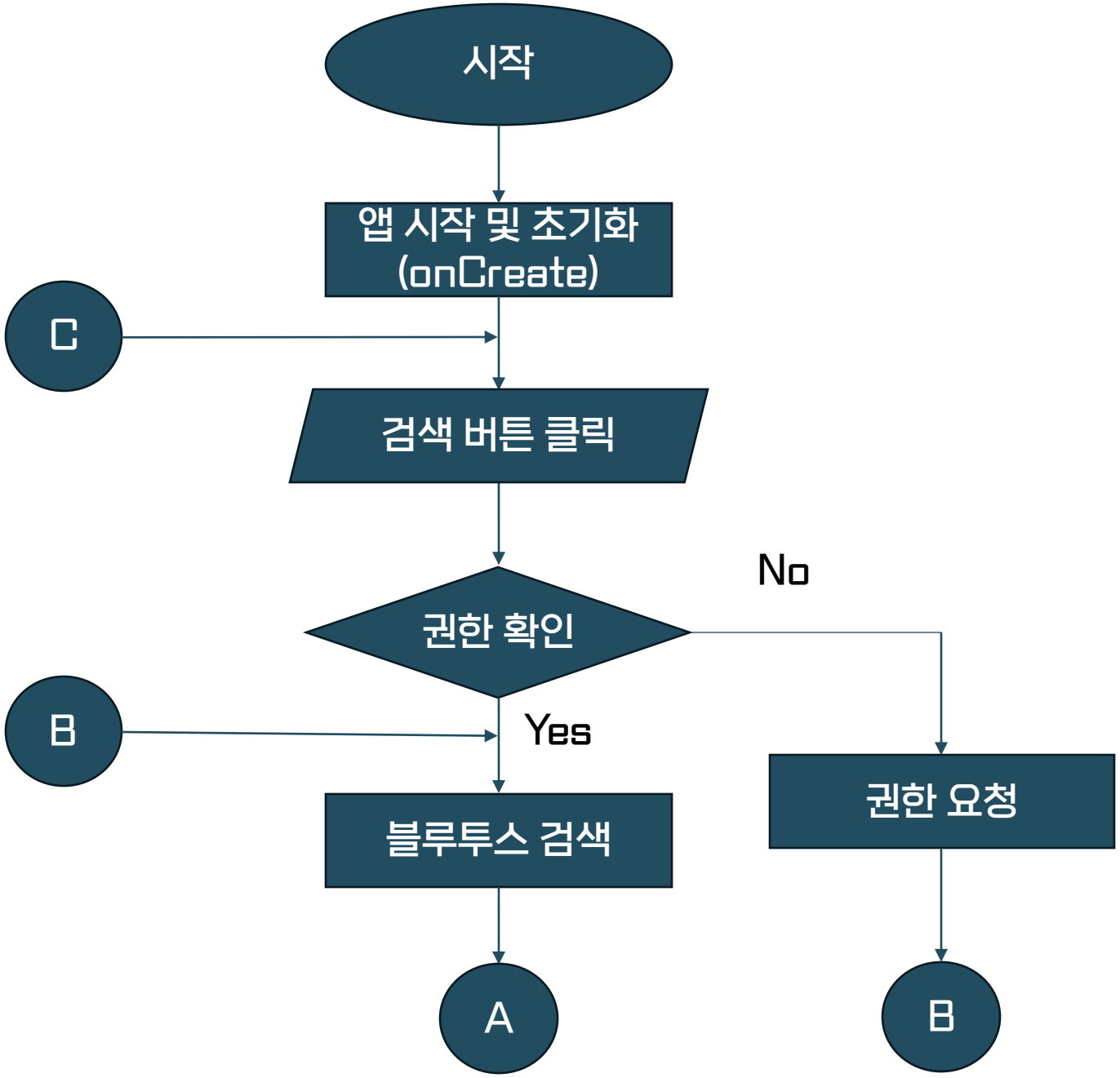
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블루투스 연결

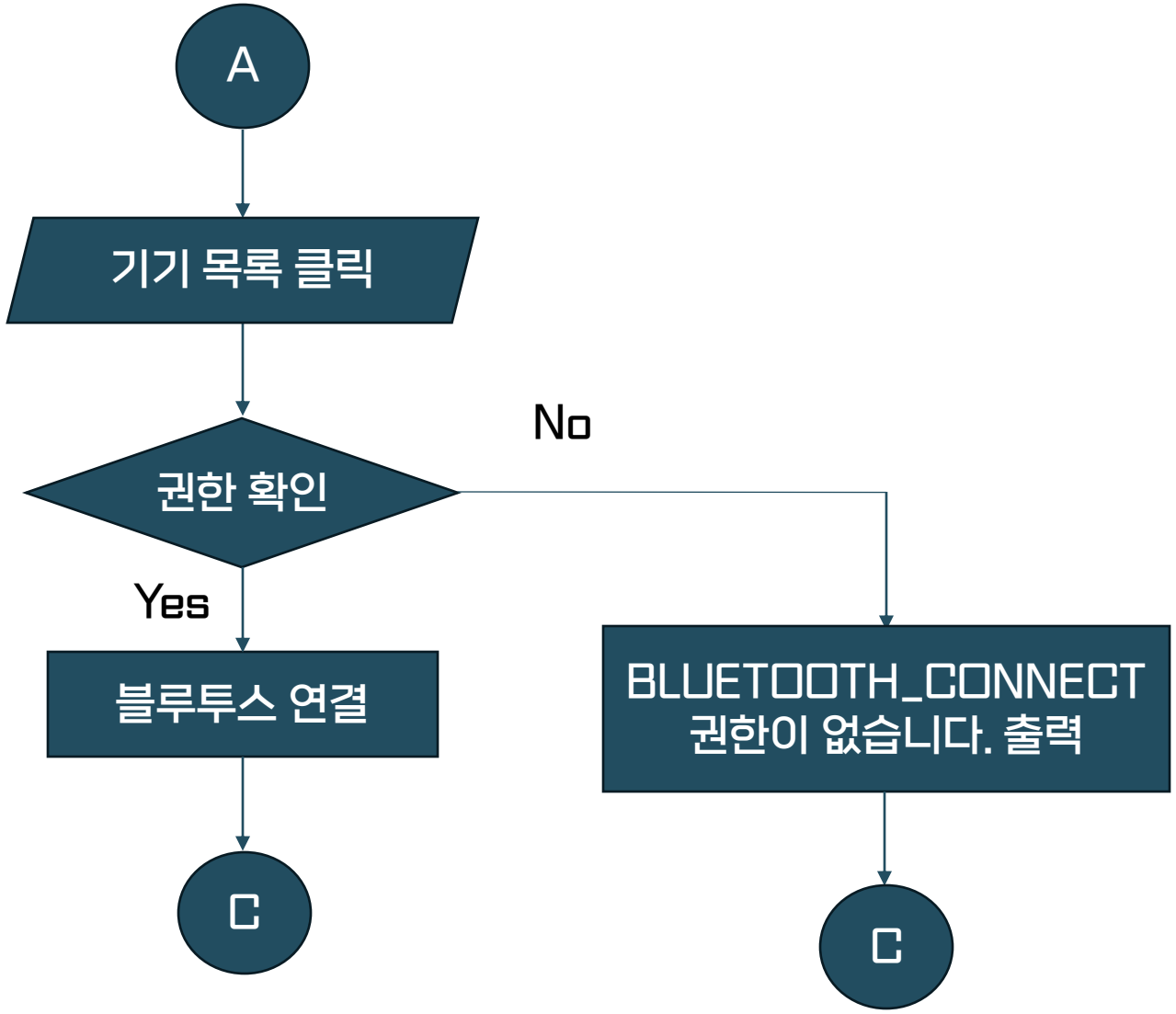
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플로우 차트



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변수 설정

```
package com.example.test1

import ...

class MainActivity : AppCompatActivity() {

    private lateinit var bluetoothAdapter: BluetoothAdapter  // lateinit으로 선언
    private val devicesList = ArrayList<BluetoothDevice>()
    private lateinit var listAdapter: ArrayAdapter<String>
    private lateinit var lvDevices: ListView
    private lateinit var btnScan: Button

    private val handler = Handler(Looper.getMainLooper())
    private var bluetoothGatt: BluetoothGatt? = null
```

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장치 발견시 처리

```
private val bluetoothReceiver = object : BroadcastReceiver() {
    override fun onReceive(context: Context?, intent: Intent?) {
        when (intent?.action) {
            BluetoothDevice.ACTION_FOUND -> {
                handleDeviceFound(intent)
            }

            BluetoothAdapter.ACTION_DISCOVERY_FINISHED -> {
                Toast.makeText( context: this@MainActivity,  text: "검색 완료", Toast.LENGTH_SHORT).show()
                btnScan.isEnabled = true
            }
        }
    }
}
```

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발견된 장치 저장

```
private fun handleDeviceFound(intent: Intent) {
    val device: BluetoothDevice? = if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        intent.getParcelableExtra(BluetoothDevice.EXTRA_DEVICE, BluetoothDevice::class.java)
    } else {
        @Suppress( ...names: "DEPRECATION")
        intent.getParcelableExtra(BluetoothDevice.EXTRA_DEVICE)
    }

    device?.let {
        if (devicesList.none { d -> d.address == it.address }) {
            devicesList.add(it)

            // 권한 체크
            val hasBluetoothPermission = if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
                ActivityCompat.checkSelfPermission(
                    context: this@MainActivity, Manifest.permission.BLUETOOTH_CONNECT
                ) == PackageManager.PERMISSION_GRANTED
            } else true
```

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발견된 장치 저장

```
            val hasLocationPermission = if (Build.VERSION.SDK_INT < Build.VERSION_CODES.S) {
                ActivityCompat.checkSelfPermission(
                    context: this@MainActivity, Manifest.permission.ACCESS_FINE_LOCATION
                ) == PackageManager.PERMISSION_GRANTED
            } else true

            val name = if (hasBluetoothPermission) it.name ?: "Unknown" else "권한 없음"
            val address = if (hasBluetoothPermission) it.address else "주소 비공개"

            if (hasLocationPermission || Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
                listAdapter.add("$name\n$address")
                listAdapter.notifyDataSetChanged()
            }
        }
    }
}
```

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권한 요청과 블루투스 확인 및 처리

```
private val requestPermissionsLauncher = registerForActivityResult(
    ActivityResultContracts.RequestMultiplePermissions()
) { permissions ->
    val allGranted = permissions.all { it.value }
    Log.d( tag: "PermissionDebug",  msg: "permissions granted: $permissions")
    if (allGranted) {
        startBluetoothDiscovery()
    } else {
        Toast.makeText( context: this,  text: "권한이 필요합니다.", Toast.LENGTH_SHORT).show()
    }
}

private val enableBtLauncher = registerForActivityResult(
    ActivityResultContracts.StartActivityForResult()
) { result ->
    if (result.resultCode == RESULT_OK) {
        startBluetoothDiscovery()
    } else {
        Toast.makeText( context: this,  text: "블루투스 활성화를 취소했습니다.", Toast.LENGTH_SHORT).show()
    }
}
```

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UI 초기화 및 이벤트 연결

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    bluetoothAdapter = (getSystemService(Context.BLUETOOTH_SERVICE) as BluetoothManager).adapter
    btnScan = findViewById(R.id.btn_scan)
    lvDevices = findViewById(R.id.lv_devices)

    listAdapter = ArrayAdapter( context: this, android.R.layout.simple_list_item_1, ArrayList())
    lvDevices.adapter = listAdapter

    btnScan.setOnClickListener {
        checkPermissionsAndScan()
    }
```

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UI 초기화 및 이벤트 연결

```
lvDevices.setOnItemClickListener { _, _, position, _ ->
    val device = devicesList[position]

    if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.BLUETOOTH_CONNECT)
        == PackageManager.PERMISSION_GRANTED
    ) {
        Toast.makeText( context: this,  text: "${device.name} 연결 시도 중...", Toast.LENGTH_SHORT).show()
        connectToDevice(device)
    } else {
        Toast.makeText( context: this,  text: "BLUETOOTH_CONNECT 권한이 없습니다.", Toast.LENGTH_SHORT).show()
    }
}

registerReceiver(bluetoothReceiver, IntentFilter().apply {
    addAction(BluetoothDevice.ACTION_FOUND)
    addAction(BluetoothAdapter.ACTION_DISCOVERY_FINISHED)
})

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v, insets ->
    val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars())
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom)
    insets
}
```

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리소스 해제

```
override fun onDestroy() {
    super.onDestroy()
    unregisterReceiver(bluetoothReceiver)

    if (ContextCompat.checkSelfPermission( context: this, Manifest.permission.BLUETOOTH_CONNECT) == PackageManager.PERMISSION_GRANTED) {
        bluetoothGatt?.close()
        bluetoothGatt = null
    } else {
        // 권한 없으면 닫지 않고 그냥 넘어감 (필요하면 로그 남겨도 됨)
        Log.w( tag: "Bluetooth",  msg: "BLUETOOTH_CONNECT permission missing, skipping bluetoothGatt.close()")
    }

    handler.removeCallbacksAndMessages( token: null)
}
```

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권한 확인

```
private fun checkPermissionsAndScan() {
    val neededPermissions = mutableListOf<String>()

    // Android 12 이상 권한
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
        if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.BLUETOOTH_SCAN) != PackageManager.PERMISSION_GRANTED) {
            neededPermissions.add(Manifest.permission.BLUETOOTH_SCAN)
        }
        if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.BLUETOOTH_CONNECT) != PackageManager.PERMISSION_GRANTED) {
            neededPermissions.add(Manifest.permission.BLUETOOTH_CONNECT)
        }
    }
```

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activity_main.xml

```
// 모든 버전에서 위치 권한 요청 (BLE 스캔에 필요)
if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED) {
    neededPermissions.add(Manifest.permission.ACCESS_FINE_LOCATION)
}

if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
    if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.POST_NOTIFICATIONS) != PackageManager.PERMISSION_GRANTED) {
        neededPermissions.add(Manifest.permission.POST_NOTIFICATIONS)
    }
}

if (neededPermissions.isNotEmpty()) {
    Log.d( tag: "PermissionDebug",  msg: "요청할 권한: $neededPermissions")
    requestPermissionsLauncher.launch(neededPermissions.toTypedArray())
} else {
    startBluetoothDiscovery()
}
```

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블루투스 검색

```
private fun startBluetoothDiscovery() {
    if (bluetoothAdapter == null) {
        Toast.makeText( context: this,  text: "블루투스를 지원하지 않는 기기입니다.", Toast.LENGTH_SHORT).show()
        return
    }

    if (!bluetoothAdapter.isEnabled) {
        val enableBtIntent = Intent(BluetoothAdapter.ACTION_REQUEST_ENABLE)
        enableBtLauncher.launch(enableBtIntent)
        return
    }

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.S) {
        if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.BLUETOOTH_SCAN) != PackageManager.PERMISSION_GRANTED) {
            Toast.makeText( context: this,  text: "BLUETOOTH_SCAN 권한이 없습니다.", Toast.LENGTH_SHORT).show()
            return
        }
    } else {
        if (ActivityCompat.checkSelfPermission( context: this, Manifest.permission.ACCESS_FINE_LOCATION) != PackageManager.PERMISSION_GRANTED) {
            Toast.makeText( context: this,  text: "위치 권한이 없습니다.", Toast.LENGTH_SHORT).show()
            return
        }
    }
```

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블루투스 검색

```
        devicesList.clear()
        listAdapter.clear()
        listAdapter.notifyDataSetChanged()

        val started = bluetoothAdapter.startDiscovery()
        if (started) {
            btnScan.isEnabled = false
            Toast.makeText( context: this,  text: "블루투스 검색 시작", Toast.LENGTH_SHORT).show()
        } else {
            Toast.makeText( context: this,  text: "블루투스 검색 시작 실패", Toast.LENGTH_SHORT).show()
        }
    }
```

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BLE 장치 연결

```
private fun connectToDevice(device: BluetoothDevice) {
    if (ContextCompat.checkSelfPermission( context: this, Manifest.permission.BLUETOOTH_CONNECT) == PackageManager.PERMISSION_GRANTED) {
        bluetoothGatt = device.connectGatt( context: this,  autoConnect: false, gattCallback)
    } else {
        // 권한 없으면 권한 요청하거나 적절히 처리
        Log.w( tag: "Bluetooth",  msg: "BLUETOOTH_CONNECT permission not granted, cannot connect to device")
        // 필요시 권한 요청 로직을 여기서 호출 가능
    }
}
```

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BLE 장치 연결

```
private val gattCallback = object : BluetoothGattCallback() {
    override fun onConnectionStateChange(gatt: BluetoothGatt?, status: Int, newState: Int) {
        super.onConnectionStateChange(gatt, status, newState)
        if (newState == BluetoothProfile.STATE_CONNECTED) {
            Log.d( tag: "BLE",  msg: "연결 성공, 서비스 탐색 시작")
            if (ContextCompat.checkSelfPermission( context: this@MainActivity, Manifest.permission.BLUETOOTH_CONNECT) == PackageManager.PERMISSION_GRANTED) {
                gatt?.discoverServices()
            } else {
                Log.w( tag: "BLE",  msg: "BLUETOOTH_CONNECT permission not granted, cannot discover services")
                // 필요 시 권한 요청이나 사용자 안내 처리 가능
            }
            runOnUiThread {
                Toast.makeText( context: this@MainActivity,  text: "연결 성공", Toast.LENGTH_SHORT).show()
            }
        } else if (newState == BluetoothProfile.STATE_DISCONNECTED) {
            Log.d( tag: "BLE",  msg: "연결 끊김")
            runOnUiThread {
                Toast.makeText( context: this@MainActivity,  text: "연결 끊김", Toast.LENGTH_SHORT).show()
            }
        }
    }
```

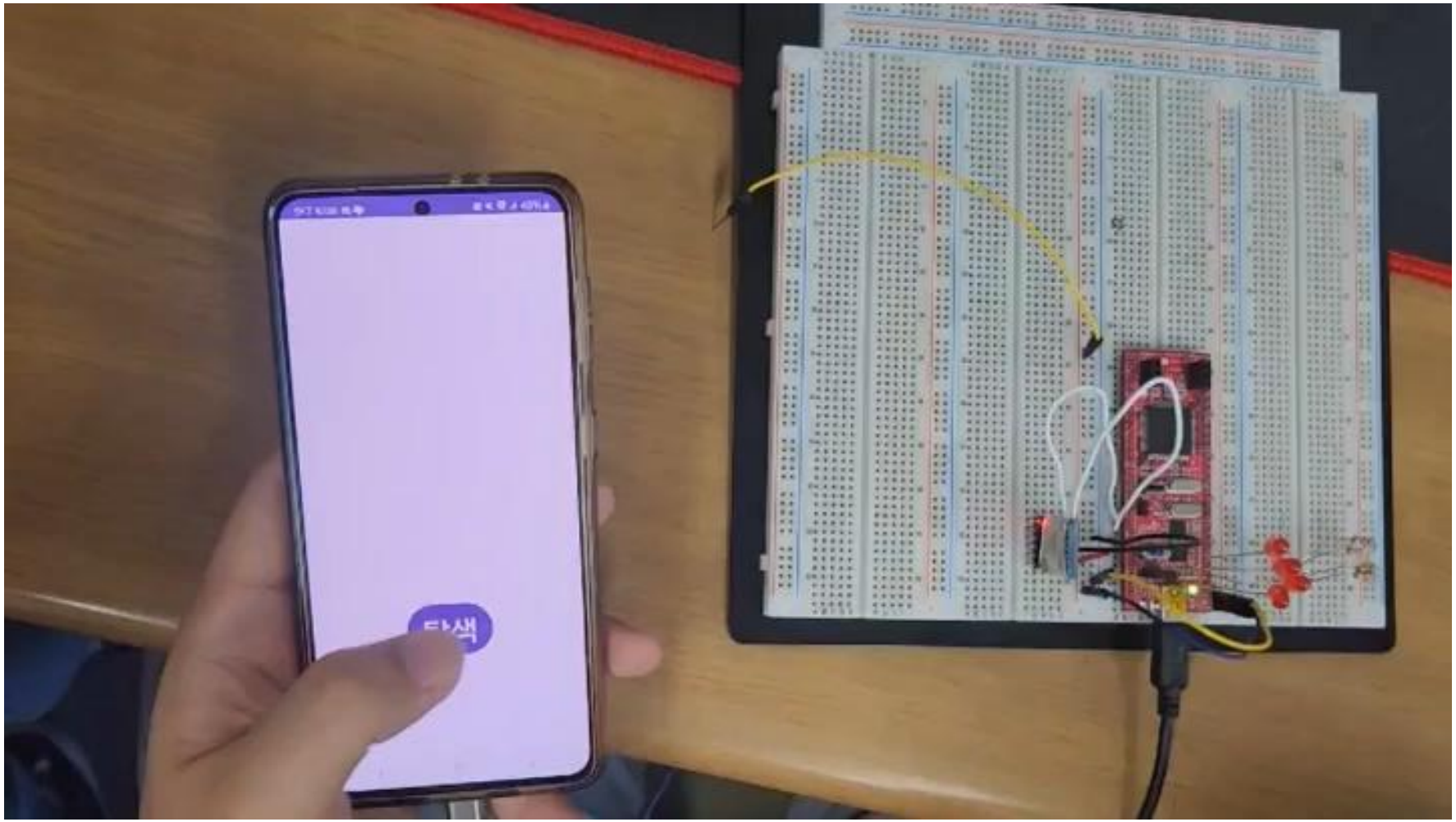
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BLE 장치 연결

```
        override fun onServicesDiscovered(gatt: BluetoothGatt?, status: Int) {
            super.onServicesDiscovered(gatt, status)
            if (status == BluetoothGatt.GATT_SUCCESS) {
                Log.d( tag: "BLE",  msg: "서비스 탐색 완료")
                // 여기서 원하는 서비스와 특성 UUID로 작업 가능
            } else {
                Log.w( tag: "BLE",  msg: "서비스 탐색 실패: $status")
            }
        }
    }
}
```

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동작



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Q&A

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